Centre Number

Candidate Number

General Certificate of Secondary Education
2019

Digital Technology

Unit 4:

Digital Development Concepts

[CODE]

SPECIMEN PAPER

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
Write your answers in the spaces provided in this question paper.
Answer all **eleven** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is **120**.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in question **3(a) and 4(a)**.

For Examiner's use only	
Question Number	Marks
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total Marks	

Examiner Only	
Marks	Re-mark

1 Insert the appropriate letter in the space provided.

(a) Debugging is:

A typing the code for a program and saving it.
B modifying the program code layout.
C examining variables within the program and checking their values for errors.
D preparing the program so that it can be understood by the computer.

_____ [1]

(b) A constant is :

A a that can be assigned but changes throughout a program.
B a that cannot be assigned and remains the same throughout the program.
C a that cannot be assigned and changes throughout the program.
D a that can be assigned but remains the same throughout the program.

_____ [1]

(c) A compiler :

A allows editing of the program code.
B performs all calculations within the program.
C executes the program.
D converts the program code to binary format.

_____ [1]

2 Complete the sentences by inserting the appropriate word from the list below.

algorithms	variables	programs

A program design solution can be developed using _____.

The designs are converted into sets of instructions for computers called _____. Locations in memory called _____ are used to hold any data being processed.

[3]

Examiner Only	
Marks	Re-mark

3 Programming solutions can be designed using abstraction and decomposition.

Quality of written communication will be assessed in this question.

(a) Explain the meaning of the terms ‘abstraction’ and ‘decomposition’.

[4]

(b) A sales person earns commission on the value of sales. The table below shows the scale of the commission.

Value Of Sales (£)	Percentage Commission
Between 1 and 999	1
Between 1000 and 1999	5
More than 2000	10

A program is required to calculate the commission and output it in a suitable format.

(i) The following variables will be used in the program. Suggest the most appropriate data types for them.

Variable	Data Type
percentageCommission	[1]
valueOfSales	[1]

A programmer has suggested that the following represents an outline solution to the problem.

Input valueOfSales
CalculateCommission
Ouput Commission

(ii) Write an algorithm for calculating the commission using the percentageCommission and the valueOfSales.

[6]

Examiner Only	
Marks	Re-mark

(iii) Testing is important when developing a solution.

Explain the following approaches to testing.

Term	Explanation
Black Box Testing	[2]
White Box Testing	[2]
Unit Testing	[2]

(iv) Design a set of test data values of valueOfSales, using the table below, which will test your solution.

valueOfSales	Reason for Test	Expected Outcome

[9]

Examiner Only	
Marks	Re-mark

4 Error trapping and validation ensure that programs work effectively for users.

Quality of written communication will be assessed in this question.

(a) Explain **three** different validation checks that are used when developing programs, **and** give an example of how each could be used.

[6]

Examiner Only	
Marks	Re-mark

(b) A pottery company supplies and delivers dishes to clients. The price depends on the type of dishes required and the amount of each dish ordered. Prices are as follows:

Type of Dish	Price
Dish Type 1 : ROUND 12 INCH MAIN COURSE PLATE	£1.00
Dish Type 2 : ROUND 10 INCH MAIN COURSE PLATE	£0.50
Dish Type 3 : SOUP BOWL	£1.50
Dish Type 4 : ROUND 6 INCH SIDE PLATE	£1.00
Dish Type 5 : OVAL MAIN COURSE DISH - ONE SIZE ONLY	£2.00

A variable called dishType has been declared. It will store the type of dish required by a client. This must be validated to ensure it is between 1 and 5.

Write an algorithm which will validate the dishType.

[4]

Examiner Only

Marks	Re-mark

Examiner Only	
Marks	Re-mark

(c) A program to output the total cost to a customer has been written. The data is entered into the program as shown below:

Client's name: Anna Greene

Type of Dish requested (1 – 5) 3

Number of Dishes required 10

The client's name will be stored as a string.

(i) List **two** string functions **and** explain how they can be used.

Function 1

Explanation 1

Function 2

Explanation 2

[4]

(ii) Explain how string functions can be used to ensure client's name is at least six characters in length and does not exceed 30 characters in length.

[2]

5 **(a)** Consider the following:

```
for i = 0 to 20
{
  if (i  %   3 == 0)
    {
      Output i
  }
}
```

In the table below, tick(✓) the line which contains the correct output from this section of code.

	Tick(✓)
A) 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20	
B) 0 3 6 9 12 15 18 21	
C) 3 6 9 12 15 18	
D) 0 3 6 9 12 15 18	
E) 3 6 9 12 15 18 21	

[1]

Examiner Only

Marks	Re-mark

Examiner Only	
Marks	Re-mark

(b) Consider the following:

```
if ((X<Y) and (Y<Z))
output(" This is the first statement")
else
if ((X<Y) or (Y>Z))
output(" This is the second statement")
else
output("This is the third statement ")
```

(i) Which statement will be executed when X =7, Y =9, Z =0?
Circle the correct letter.

A This is the first statement
B This is the second statement
C This is the third statement [1]

(ii) Which statement will be executed when X=12, Y=12, Z=18?
Circle the correct letter.

A This is the first statement
B This is the second statement
C This is the third statement [1]

6 A program is required which will take in 10 values in pounds and output the corresponding kilogram value. The pounds will be stored in an ARRAY called pounds and the calculated kilogram values will be stored in an array called KILOS. One pound equals 0.4545 kilos. So for example:

pounds[0]=1.0 kilos[0]=0.4545
pounds[1]=2.0 kilos[1]=0.9090

(a) Declare an array called pounds which will hold 10 integer values.

_____

_____

_____

_____

_____

_____ [3]

(b) Write an algorithm to initialize the contents of the pounds array to zero.

[3]

(c) Write the code required to read in the values in pounds into the array from the user screen.

[3]

7 **(a)** A procedural program uses sequence, selection and iteration.

In the table below circle to indicate whether the operations use a sequence, selection or iteration.

Operations			
Performing a series of operations one after the other.	Sequence	Selection	Iteration [1]
The calculation of an employee's tax payment depends on their salary level.	Sequence	Selection	Iteration [1]
The program will continue to add the cost of a basket of items until the basket is empty.	Sequence	Selection	Iteration [1]

Examiner Only

Marks	Re-mark

(b) State what is meant by each of the following terms used in Procedural programming:

(i) Global variable

_____

_____

_____ [1]

(ii) Local variable

_____

_____

_____ [2]

(c) State what is meant by each of the following terms in Object-oriented programming:

(i) Class

_____

_____

_____

_____ [2]

(ii) Object

_____

_____

_____

_____ [2]

Examiner Only	
Marks	Re-mark

(d) A programmer can use an Integrated Development Environment (IDE) to create and develop a program.

Describe the facilities available in an integrated development environment (IDE) which can help a programmer to identify and correct syntax errors.

[4]

8 **(a)** State what is meant by the character set of a computer.

[1]

(b) Explain how ASCII is used to represent text in a computer system.

[3]

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Marks	Re-mark

(c) Unicode is also used to represent text in a computer system.

Explain the difference between the character sets of Unicode and ASCII.

_____

_____

_____

_____ [2]

9 **(a)** **(i)** Convert the denary number 36 into a binary number.
(Working out must be clearly shown.)

_____

_____ [2]

(ii) Convert the denary number 22 into a hexadecimal number
(Working out must be clearly shown.)

_____

_____ [3]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

(iii) Explain why hexadecimal numbers are often used to represent binary numbers.

[2]

(b) (i) Add the following bytes:

```
  0 1 1 0 0 0 1 0
+ 1 1 0 1 0 1 0 0
-----------------
```

[2]

(ii) Explain the problem that will occur if the result of the addition is stored as a byte.

[2]

(c) Complete the following truth tables:

OR Gate		
A	**B**	**X**
0	0	0
0	1	
1	0	
1	1	1

[1]

[1]

AND Gate		
A	**B**	**X**
0	0	0
0	1	
1	0	
1	1	1

[1]

[1]

[4]

10 The following numbers represent pupil heights in cm. They are stored in an array called studentHeight. The content of the array is to be sorted in ascending order.

155,152,156,151,153

(a) (i) Name and describe a sorting technique which could be used to sort the numbers. Include information about:

- the loop structures used; and
- the efficiency of the sort.

Name: _____

Loop structures used

_____

_____

_____

Efficiency of the sort

_____

_____

_____

[5]

Examiner Only	
Marks	Re-mark

(ii) Write the condition which will be used in the 'if' statement for a sort process.

_____

_____

_____

_____ [2]

(iii) For the method chosen in **(i)** on the previous page, demonstrate how the data in studentHeight would be organised during the sorting process. Show clearly the content on the array after each pass.

studentHeight

155	152	156	151	153

Pass 1

Pass 2

Pass 3

Pass 4

Pass 5

[5]

(b) Once an array has been sorted it can be searched efficiently.

Describe a method of searching which is more efficient than linear search.

_____

_____

_____

_____

_____

_____ [3]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

11 A program is required to take in three numbers and output the average and product of the three numbers. A top level algorithm could be:

DISPLAY INTRO SCREEN
GET 3 NUMBERS
FIND AVERAGE OF 3 NUMBERS
FIND PRODUCT OF 3 NUMBERS
OUTPUT INFORMATION

User-defined functions could be written for each of the above.

(a) (i) What is a user-defined function?

[3]

(ii) Write a function called **findAverage** which will calculate the average of **three** integers.

[3]

(iii) Write a function called **findProduct** which will calculate the product of **three** integers.

Examiner Only	
Marks	Re-mark

[3]